Illinois Property & Casualty Market Share Analysis

An End-to-End Modern Data Stack Case Study

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Abstract

This report details the technical architecture and business insights derived from the 2024 Illinois Property & Casualty (P&C) insurance market data. By leveraging a modern data stack consisting of Snowflake, dbt, and Power BI, this project successfully transformed unstructured regulatory filings into a high-fidelity executive dashboard, identifying key market leaders and competitive dynamics.

1 Project Introduction

The objective of this analysis was to provide a "Source of Truth" for insurance stakeholders in Illinois. The primary challenge involved extracting value from static regulatory reports and ensuring that complex calculations, such as market share percentages, remained accurate across various levels of data aggregation.

2 The Modern Data Stack

The project was built using a three-tier architecture to ensure scalability and data integrity:

- **Data Warehouse (Snowflake):** Served as the centralized repository. Raw data from PDF extractions was ingested into localized tables, maintaining a historical record of regulatory filings.
- **Transformation Layer (dbt):** Utilized for data modeling. I implemented a Medallion Architecture (Staging → Gold) where raw premiums were cleaned and market share calculations were performed using SQL window functions.
- **Visualization (Power BI):** Developed an interactive dashboard to communicate 5.7 billion dollars in market volume through intuitive visual hierarchies.

3 Technical Challenges & Solutions

3.1 Data Normalization

Data originally sourced from PDF format contained inconsistencies in company naming (e.g., abbreviations vs. full legal names). Using SQL transformations, I standardized the `COMPANY_NAME` field to ensure that distinct counts (222 unique insurers) were accurate.

3.2 Market Share Logic

A significant technical hurdle was ensuring the Donut Chart and Treemap accurately reflected market share when filters were applied.

- **Initial Issue:** Filtering to a "Top 10" view caused the market leader (State Farm) to appear as having 44% share because the visual was calculating share based on the filtered subset.
- **Solution:** Developed a robust measure ensuring percentages were always calculated against the **Total Market Volume ($5.7bn)**, resulting in a consistent and accurate **28.73%** share for the market leader.

4 Key Business Insights

- **Market Scale:** The 2024 Illinois P&C market represents **$5.68 Billion** in direct written premiums.
- **Dominance: State Farm Mut Auto Ins Co** remains the clear market leader with a **28.73%** market share, followed by Progressive Universal (8.08%) and Allstate Fire & Cas (7.81%).
- **Fragmentation:** Outside of the top 10 players, there is a significant "long tail" of over 200 insurers, representing a competitive and fragmented secondary market.

5 Conclusion

This project demonstrates the power of combining engineering rigor with business intelligence. By moving data from static documents to a dynamic warehouse and dashboard, we have enabled stakeholders to perform real-time competitive analysis with verified, high-integrity data.

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